MANIPAL INSTITUTE OF TECHNOLOGY

RL-Driven Adaptive Data Augmentation for Traffic Sign Recognition

Using CNNs and YOLO-Based Cropping

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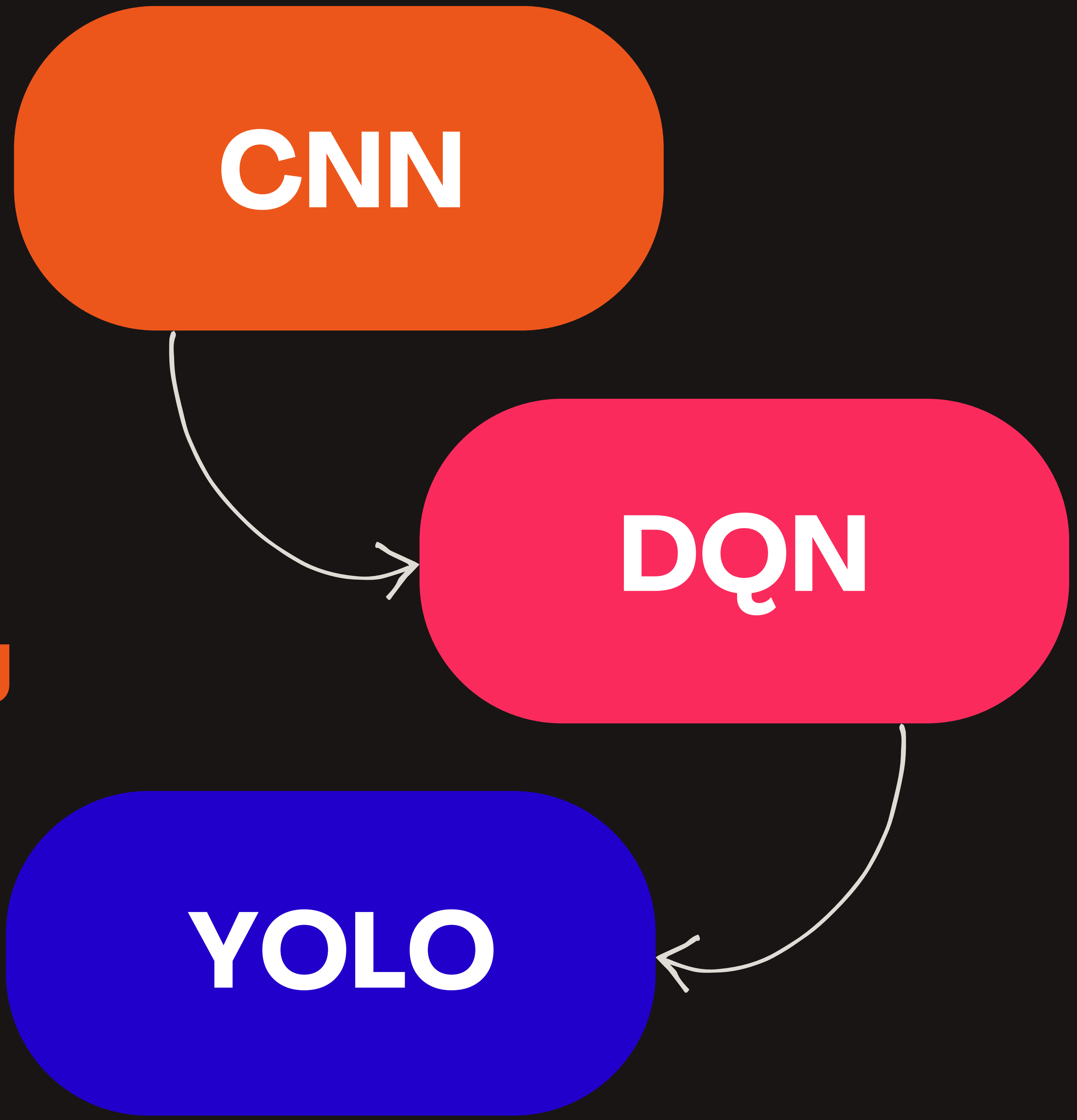


Introduction & Motivation

THE PROBLEM

- Traffic sign recognition is safety-critical for autonomous vehicles
- Real-world conditions degrade CNN performance: illumination shifts, occlusions, scale variation
- Static augmentation pipelines apply fixed transformations regardless of where the model struggles

THE DEPLOYMENT GAP

- GTSRB images are pre-cropped and neatly framed from German roads
- Real dashcam footage has motion blur, messy backgrounds, variable scales
- Models trained on benchmarks may fail catastrophically in deployment

43

sign categories in GTSRB with heavily skewed class frequencies

Misread signs in autonomous driving can have severe real-world consequences

The Proposed Solution

TWO KEY INNOVATIONS

01 RL-Driven Augmentation

- Deep Q-Network (DQN) agent learns optimal augmentation policies dynamically
- State: validation accuracy, loss, previous action (8-dimensional vector)
- Action: 6 augmentation operations selected per epoch
- Reward: per-epoch change in validation accuracy — directly optimizes generalization

02 YOLO-Based Crop Extraction

- YOLOv8 pipeline extracts traffic sign crops from real-world dashcam footage
- Bridges the gap between benchmark datasets and practical deployment
- Every 10th frame extracted → YOLOv8 inference → crop, pad, resize to 64×64
- Enables pseudo-labelling for future training integration

Dataset: GTSRB & Preprocessing

39,209
Training Images

12,630
Test Images

43
Sign Categories

<200 to >2000
Sample Range per Class

DATASET CHALLENGES

- Heavily skewed class frequencies — minority classes have fewer than 200 samples
- Image sizes vary from 15×15 to 250×250 pixels
- Quality degraded by motion blur, variable illumination, partial occlusion

PREPROCESSING STEPS

- Images resized to 32×32 px via bilinear interpolation, rescaled to [0,1]
- Stratified 85%/15% train/validation split to maintain minority class proportions
- No per-channel normalization; global rescaling sufficient empirically

Methodology: CNN Architecture

Input
32×32×3

Conv2D(32)
3×3 ReLU
MaxPool 2×2

Conv2D(64)
3×3 ReLU
MaxPool 2×2

Flatten
Dense(128)
ReLU Drop(0.3)

Dense(43)
Softmax

TRAINING CONFIGURATION

Optimizer	Loss Function	Baseline Epochs	Fixed Aug Epochs	RL Epochs
Adam (lr = 10^{-3})	Sparse Categorical Cross-Entropy	20 (no callbacks)	30 (early stopping, patience=7)	50 (full RL loop)

Methodology: RL-Driven Augmentation (DQN)

STATE SPACE (dim=8)

- Current validation accuracy
- Current validation loss
- One-hot encoding of previous action (6-dim)

8-dimensional vector observed after each epoch

ACTION SPACE (6 ops)

- Rotation (±15°)
- Brightness (±0.3)
- Gaussian Noise (σ=0.05)
- Translation (10% dims)
- Zoom (−15% to +15%)
- Identity (no aug)

One action selected per training epoch

REWARD SIGNAL

- r_t = val_acc_t − val_acc_t−1
- Incentivizes augmentations that improve generalization
- No shaping or discount beyond Bellman backup

Directly optimizes the metric that matters

DQN: 3-layer MLP (8→64→64→6) | MSE loss | Adam (lr=10^{-3}) | Replay buffer: 2,000 | Target net sync: every 5 steps | ε-greedy: 1.0 → 0.10 (decay 0.97/step)

Methodology: YOLO-Based Crop Extraction

PURPOSE: Bridge the train/inference gap by generating supplementary data that mirrors real deployment conditions.

Frame Extraction
Extract every 10th frame from dashcam footage.

1

YOLOv8 Inference
Detect traffic signs on full frames using YOLOv8.

2

Crop & Pad
Add 8-pixel padding around each detected bounding box.

3

Resize & Filter
Resize to 64×64 and discard crops smaller than 20×20 px.

4

WHAT IT PRODUCES

Real-world crops with motion blur, messier backgrounds, and variable scale — substantially different from clean GTSRB images.

FUTURE DIRECTION

Pseudo-labelling at 0.85 confidence threshold: route high-confidence predictions to training, uncertain crops to manual review.

Results: Comparative Performance

98.34%

Best Test Accuracy

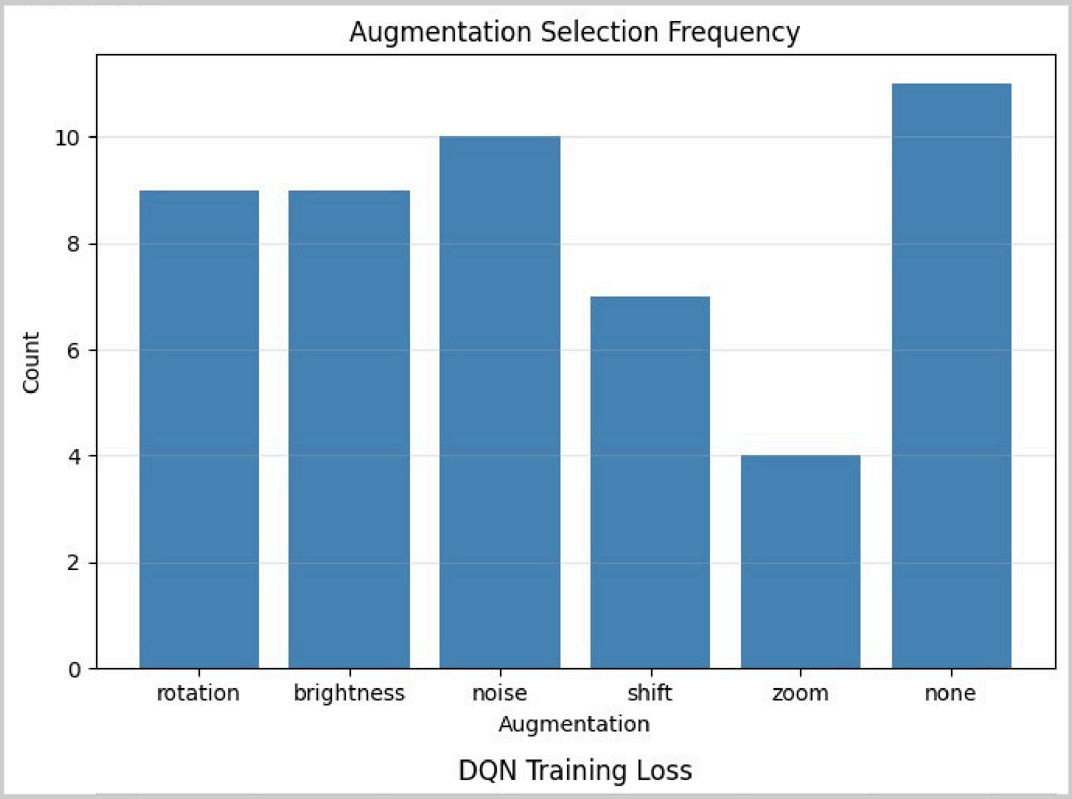
~75 fewer misclassifications vs fixed aug

Method	Test Accuracy (%)	Epochs
Baseline CNN (no augmentation)	96.33	20
Fixed Augmentation	97.74	30
RL-Driven Augmentation (DQN)	**98.34 ★**	**50**

ACCURACY GAINS

Analysis: DQN Diagnostics

Augmentation Selection Frequency

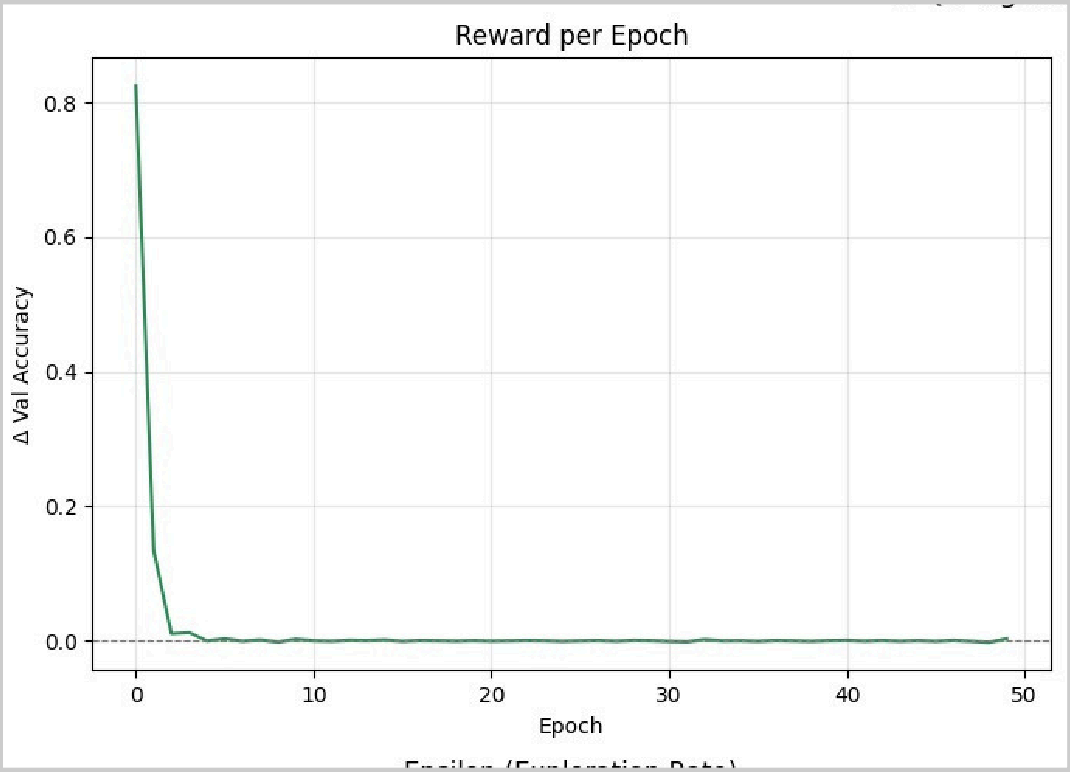


Identity selected most (11×), Noise second (10×). Zoom least (4×).

Agent learned when to leave the model alone — crucial insight.

Identity +Noise = 42%

Reward per Epoch

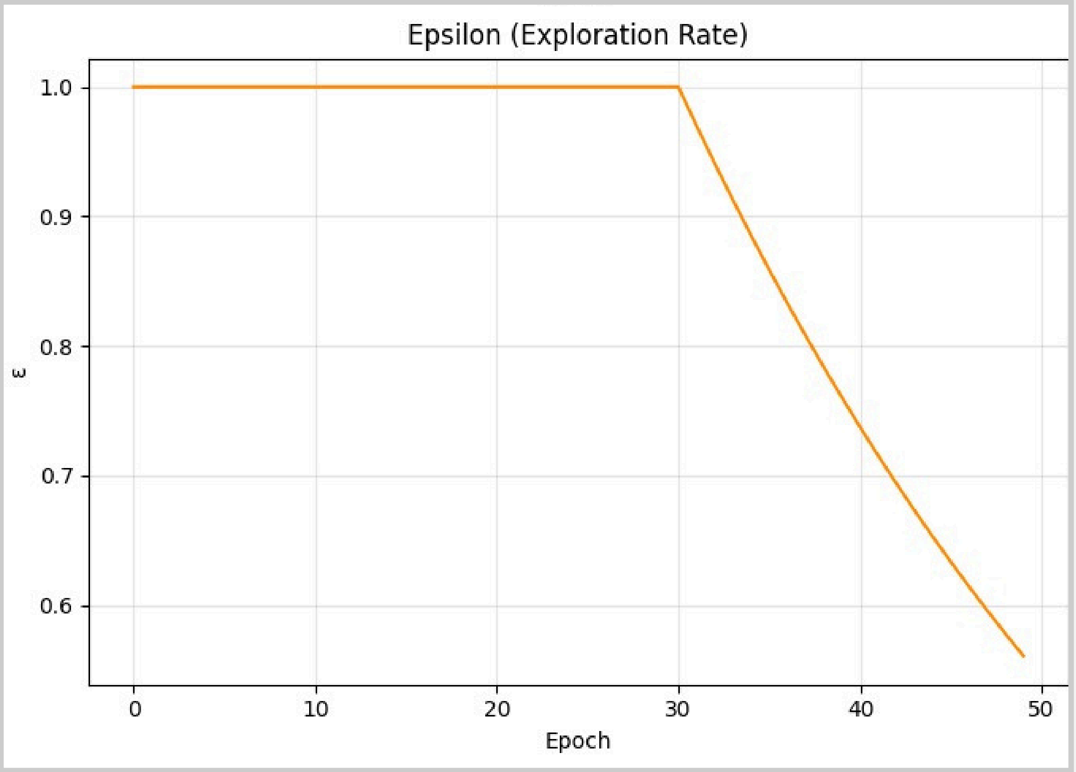


Peak reward +0.83 at epoch 1 (largest single-epoch accuracy jump).

Rapid decrease as model saturates — reward signal becomes sparse.

Peak +0.83

Epsilon Decay & Exploration



Epochs 1–15: high oscillation in training accuracy (ε ≈ 1.0, random aug).

After epoch 30: ε decays, training stabilizes as policy is exploited.

Val Acc >99% from ep.5

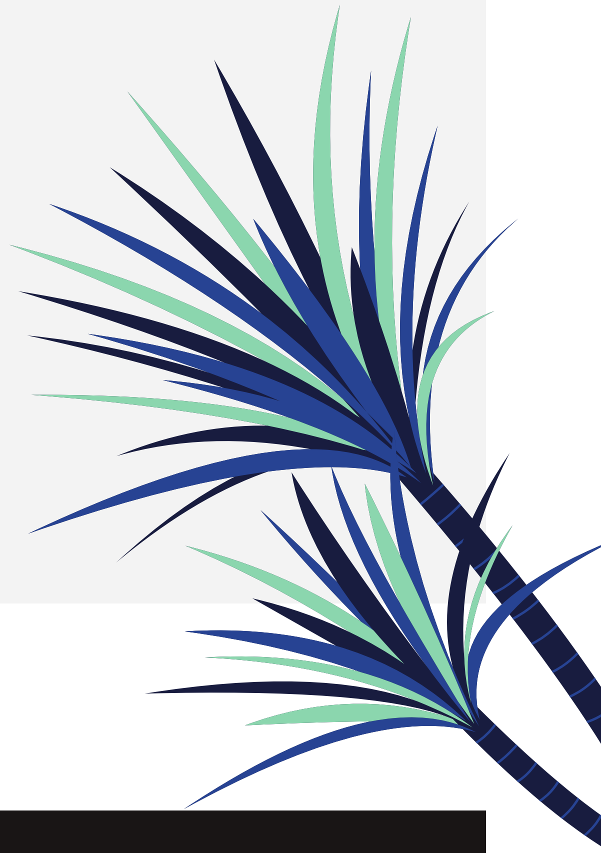
Discussion: Deployment Gap & Limitations

THE DEPLOYMENT GAP

- **GTSRB is a clean benchmark** – real dashcam footage is significantly harder
- **Real footage**: more pronounced motion blur, messier backgrounds, arbitrary sign scales and positions
- **YOLO pipeline makes this gap explicit and measurable** – critical step for deployment readiness

CURRENT LIMITATIONS

- DQN spent first ~30 epochs in near-pure exploration ($\varepsilon \approx 1.0$) – limited early influence on training
- The 0.60-point gain over fixed augmentation is modest, partly due to GTSRB's clean nature
- Action space limited to 6 operations; more diverse space could reveal larger RL advantage



FUTURE WORK: Tighter ε decay schedule • Pre-loaded replay buffer with known-useful transitions • Pseudo-labelling YOLO crops at 0.85 confidence • Expanded action space

Conclusion

1

Adaptive > Static

RL-driven augmentation achieves **98.34%** — demonstrating that dynamically responding to model feedback outperforms fixed augmentation schedules.

2

Agent Self-Awareness

The DQN effectively **learned when to apply augmentation** and when to leave the model alone — choosing Identity in 22% of epochs near saturation.

3

Real-World Readiness

Integrating object detection pipelines like YOLOv8 is essential for **bridging the gap** between benchmark accuracy and real deployment conditions.

A promising direction for more robust and adaptable AI in autonomous driving.

KEY REFERENCES

[1] **Stallkamp et al.** – Man vs. Computer: Benchmarking ML for Traffic Sign Recognition. Neural Netw., 2012.

[2] **Sermanet & LeCun** – Traffic Sign Recognition with Multi-Scale CNNs. IJCNN, 2011.

[3] **Ciresan et al.** – Multi-Column Deep Neural Network. Neural Netw., 2012.

[4] **Cubuk et al.** – AutoAugment: Learning Augmentation Policies from Data. CVPR, 2019.

[5] **Cubuk et al.** – RandAugment: Practical Automated Data Augmentation. CVPRW, 2020.

[6] **Jocher et al.** – Ultralytics YOLOv8. 2023.

Q&A

Questions Welcome

Google Colab Available

All models, code, and results available for detailed review upon request.